Assignment 5 Name Logan Langford

1. Your target for this assignment is to create a web BMI and Calorie calculator that lets the user enter their weight (in kg), height (in cm), age and gender, then calculate their Body Mass Index (BMI) and basic daily calories needed. It will then generate a suggested meal plan from a provided database that fits their daily calorie needs.
2. Formulas:
   a. BMI = weight / (height/100)^2
   b. Basic Calories
      i. Men: (weight x 10) + (height x 6.25) - (age x 5) + 5
      ii. Women: (weight x 10) + (height x 6.25) - (age x 5) -161
   c. Calorie Target
      i. If Obese (BMI >= 30.0), subtract 750 from Basic Calories.
      ii. If Overweight (BMI 25.0 to 29.9), subtract 500 from Basic Calories.
      iii. If Normal/Underweight (BMI < 25.0), use Basic Calories.
3. Application Requirement

- ☐ Name your file healthplan.html
- ☐ In the assignment section of your index.html document, create a link which links to this document "healthplan.html".
- ☐ Create an interface with input text boxes and labels for Weight (kg), Height (cm), and Age.
- ☐ Create a Drop-down list (<select> tag) for Gender (Men/Women).
- ☐ Display the calculated BMI (rounded to two decimal places) and their Calorie Target.
- ☐ Meal Plan: Design a JavaScript function that selects food from the foodDatabase array given. The total calories of the selected foods must be as close to the Calorie Target as possible without going over. Display the final list of foods and the total meal plan of calories on the screen. When the user clicks "generate meal plan" button again, a brand-new set of food choices will be displayed.
- ☐ Algorithm Explanation: Explain the logic you used to generate the meal plan. How does your loop decide which foods to pick, and how does it ensure the total stays under the Calorie Target? (Do not just paste your code—explain the *concept. **Use only if/else if to implement your logic **)*

In my code I use 2 for loops to get the meal plan based on the user's calories. The first loop uses the calorie target to generate what calories can fit within the limit/calorie target. The second loop randomizes the the food names

- ☐ Animation: Create an animation that dynamically changes the width/height of the image of a person so that it visually represents the calculated BMI.
- ☐ Syntax Restriction: You are not allowed to use inline HTML event handlers, **.map(), .filter(),.reduce(), .find()** and other advance array methods. Use **only** fundamental methods taught in class.
- ☐ Before you submit your file, be sure to check this checklist.
- ☐ To submit:
  - ○ Upload your files to GitHub.
  - ○ Please turn in this page with items checked to instructor on the due date.